

Local Projections or VARs: Evidence from Monte-Carlo Simulations

Aaron Liebig & Patrick Tenner

Superv.: Prof. Dr. Derya Uysal

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Motivation

- VARs and Local Projection used for Impulse Response Analysis, think Unemployment and Interest Rate, Oil Prices and Inflation, ...
- In population both estimate the same Impulse Response
- finite sample comparison ($\text{VAR}(p)$ vs. $\text{LP}(p)$)
 1. In finite samples, bias-variance puzzle
 2. VARs more efficient, LPs more robust to misspecification
- Ludwig identity shows that equal lag comparison is inherently flawed

(BL:) Where does a fixed $\text{LP}(4)$ sit on the VAR complexity frontier and does the conventional equal-lag ranking survive?

(EX:) Does any genuine LP advantage emerge once misspecification is introduced?

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Agenda

1. Theory
2. Study Design
3. Results
4. Discussion
5. Conclusion

VARs

$$Y_t = \nu + \mathbf{A}Y_{t-1} + U_t \xrightarrow[\text{restr. } K]{A1} y_t = \mu + \sum_{\ell=0}^{\infty} \Psi_{\ell} u_{t-\ell}, \quad \Psi_{\ell} := [\mathbf{A}^{\ell}] \quad (1)$$

Identification: As $\sum_u = \mathbb{E}[u_t u_t'] \neq I_K$, Define $u_t = B\varepsilon_t$ where $\varepsilon_t \sim (0, I_k)$

$$y_{t+h} = \mu + \sum_{\ell=0}^{\infty} \Theta_{\ell} \varepsilon_{t+h-\ell} \quad (2)$$

Suffer from

- LS bias compounds as horizon grows Kilian 1998
- near-unit root behavior
- misspecification through MA dynamics or lag order

LPs

$$y_{t+h} = \alpha_h + \beta_h s_t + \gamma'_h \mathbf{x}_t + \xi_{t+h}^{(h)}, \quad h = 0, 1, \dots, H, \quad (3)$$

Equivalence and Ludwig identity

DGP's

Method

Baseline Results - Complexity Frontier

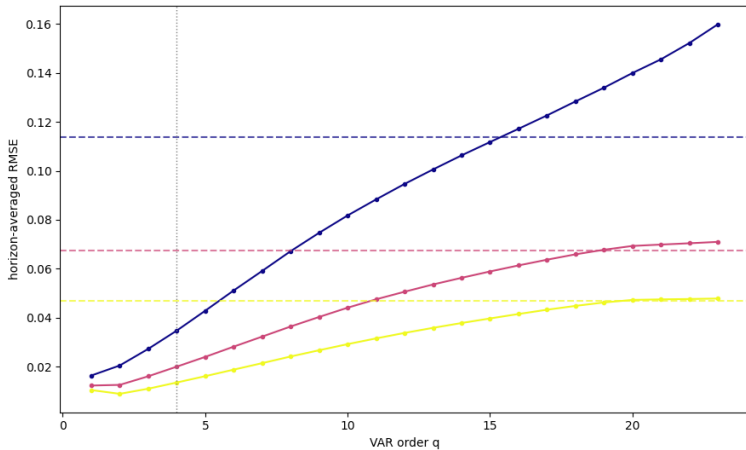


Figure 1: Complexity Frontier $\rho = 0.5$

Complexity Frontier - varying ρ

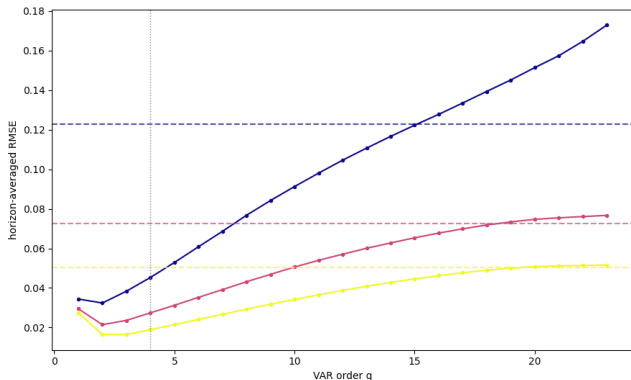


Figure 2: Complexity Frontier $\rho = 0.7$

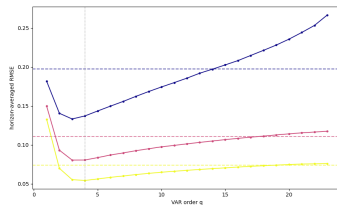


Figure 3: Complexity Frontier $\rho = 0.95$

Complexity Frontier - varying ρ

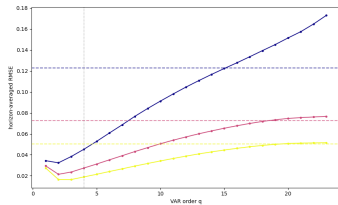


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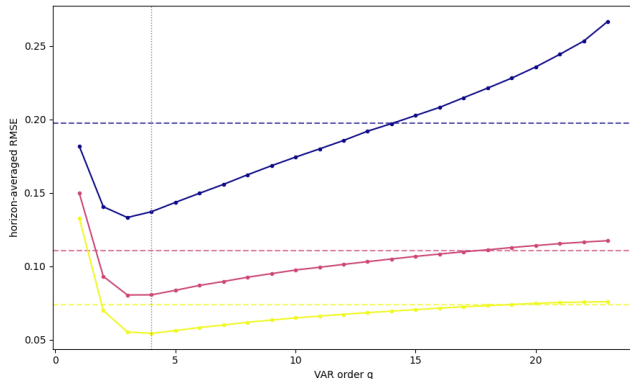


Figure 3: Complexity Frontier $\rho = 0.95$

Bias

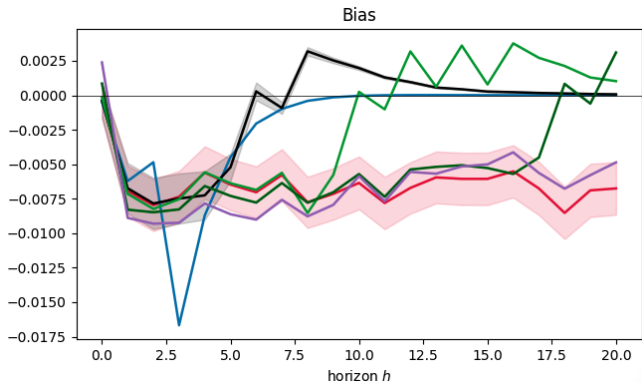


Figure 4: Bias at $\rho = 0.5$

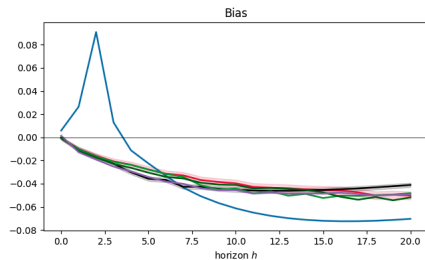


Figure 5: Bias at $\rho = 0.95$

Bias

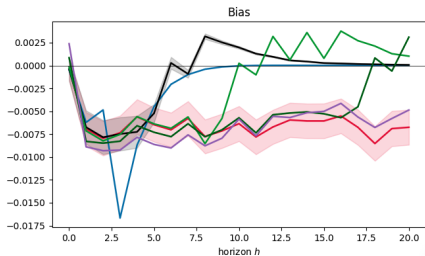


Figure 4: Bias at $\rho = 0.5$

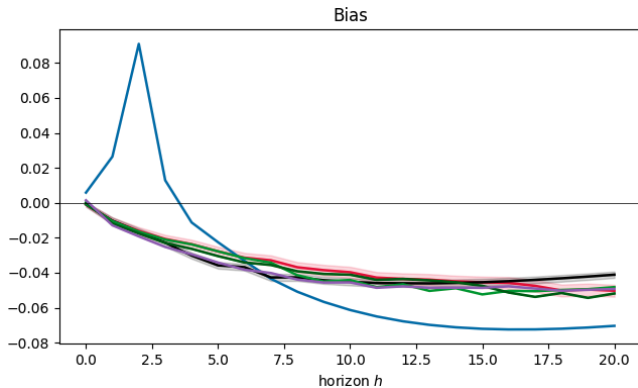


Figure 5: Bias at $\rho = 0.95$

Variance

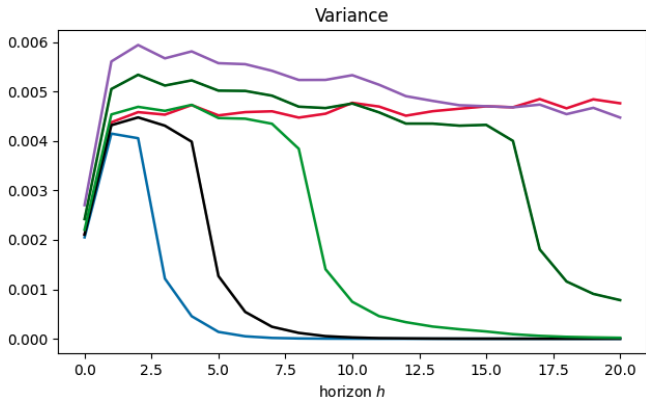


Figure 6: Variance at $\rho = 0.5$

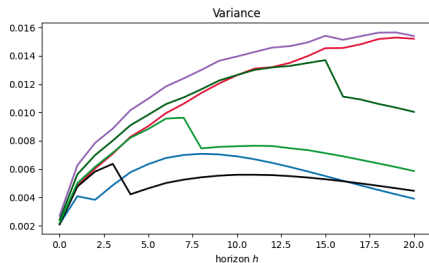


Figure 7: Variance at $\rho = 0.95$

Variance

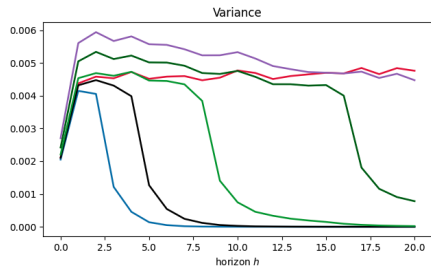


Figure 6: Variance at $\rho = 0.5$

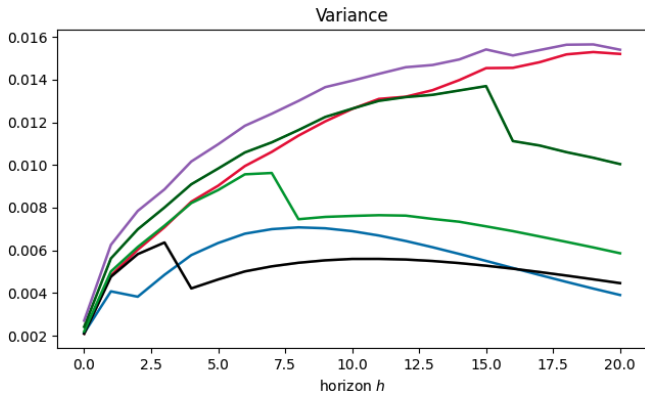


Figure 7: Variance at $\rho = 0.95$

Coverage

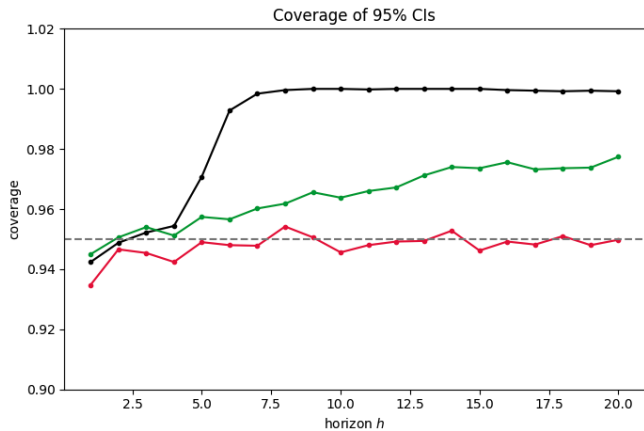


Figure 8: Coverage at $\rho = 0.5$

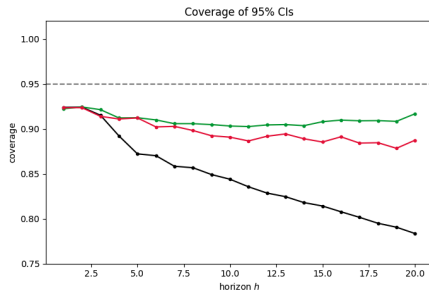


Figure 9: Coverage at $\rho = 0.95$

Coverage

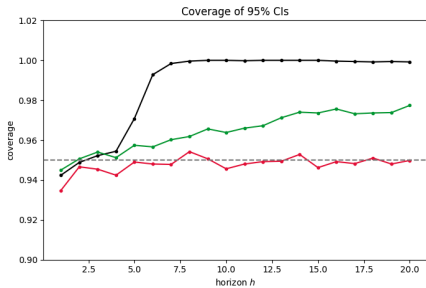


Figure 8: Coverage at $\rho = 0.5$

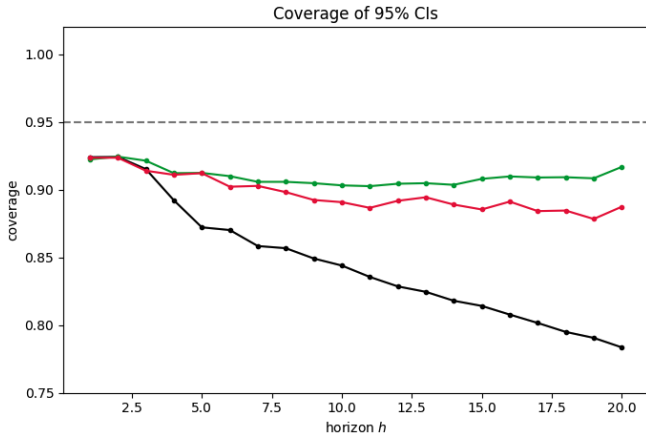


Figure 9: Coverage at $\rho = 0.95$

Introducing Dynamic Misspecification

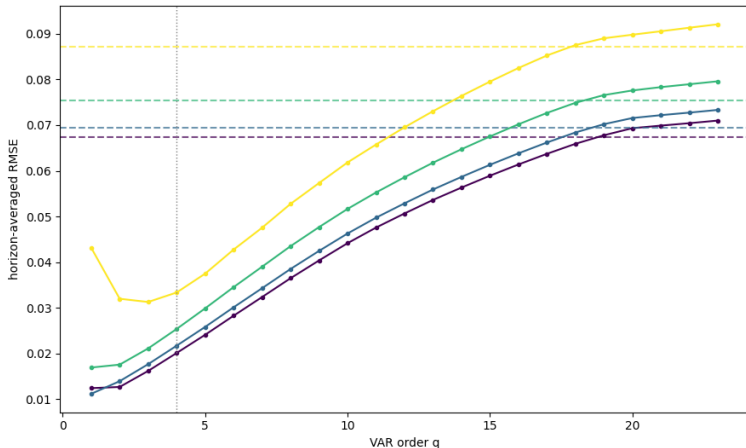


Figure 10: Complexity Frontier $\rho = 0.5$

Dynamic Misspecification - varying ρ

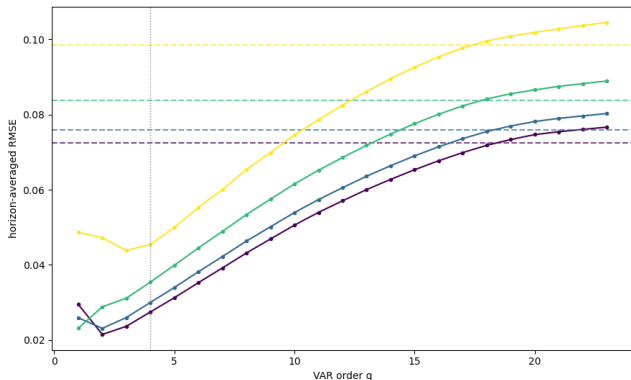


Figure 11: Complexity Frontier $\rho = 0.7$

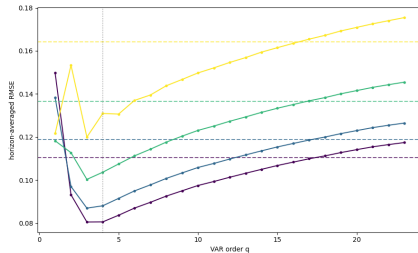


Figure 12: Complexity Frontier $\rho = 0.95$

Dynamic Misspecification - varying ρ

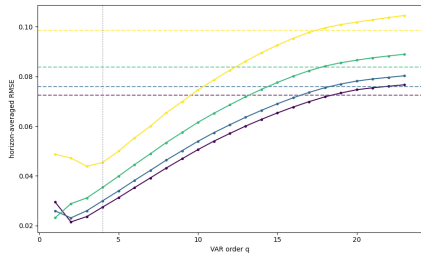


Figure 11: Complexity Frontier $\rho = 0.7$

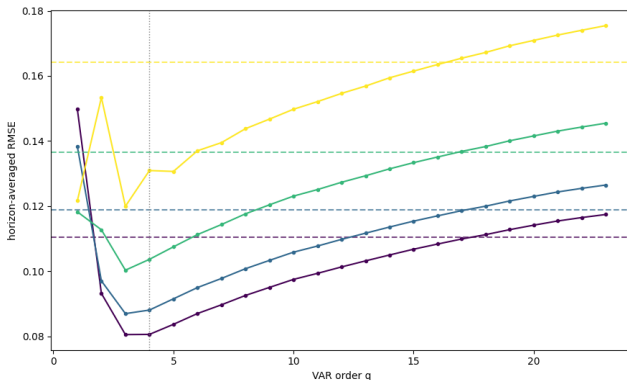


Figure 12: Complexity Frontier $\rho = 0.95$

Bias - Correct vs. Misspecification - $\tau = 0.6$

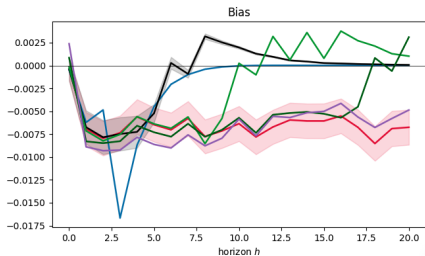


Figure 4: Baseline at $\rho = 0.5$

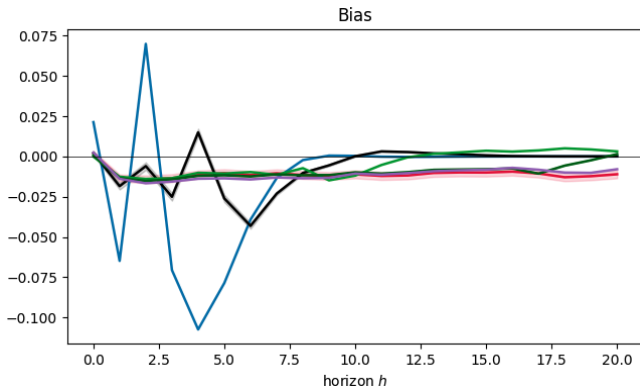


Figure 13: Dyn. Misspecification at $\tau = 0.6$

Variance - Correct vs. Misspecification - $\tau = 0.6$

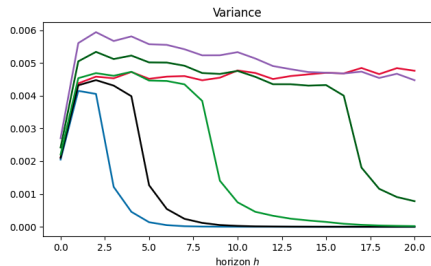


Figure 6: Baseline at $\rho = 0.5$

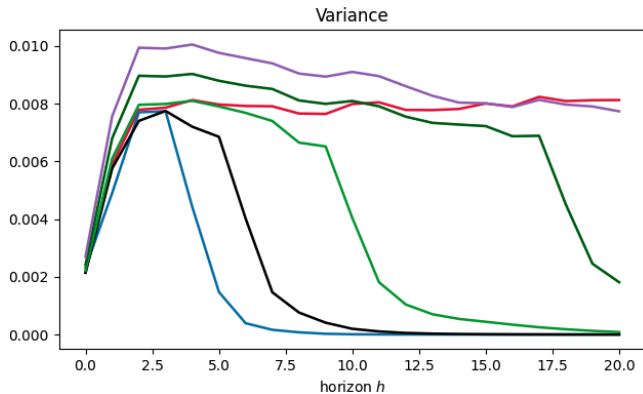


Figure 14: Dyn. Misspecification at $\tau = 0.6$

Coverage - Correct vs. Misspecification - $\tau = 0.6$

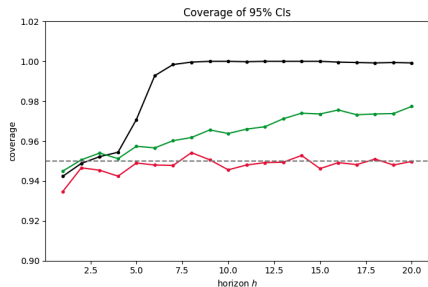


Figure 8: Baseline at $\rho = 0.5$

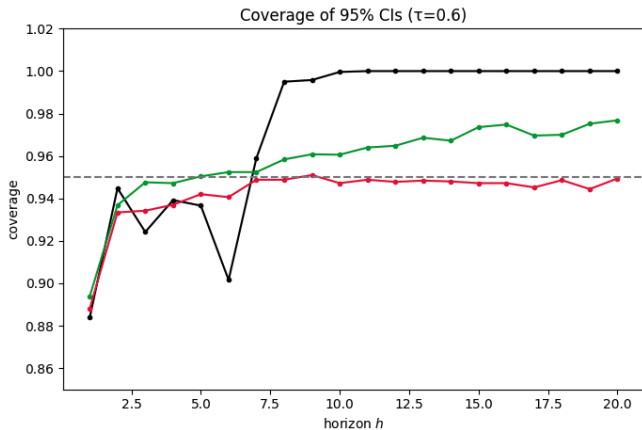


Figure 15: Dyn. Misspecification at $\tau = 0.6$

Introducing Functional Misspecification

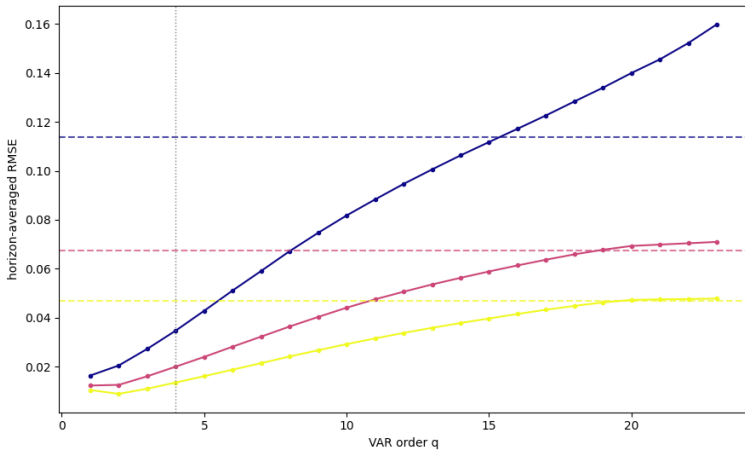


Figure 16:

Variance - Correct vs. Misspecification - $\rho = 0.5$

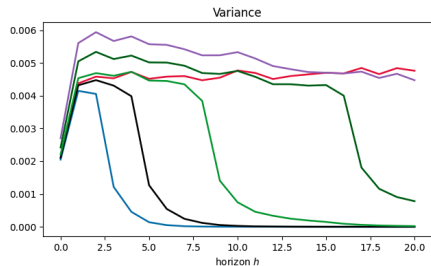


Figure 6: Baseline at $\rho = 0.5$

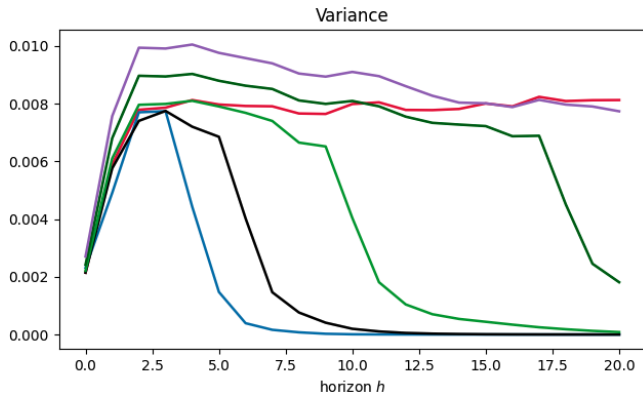


Figure 14: Dyn. Misspecification at $\gamma = 0.6$

Coverage - Correct vs. Misspecification - $\rho = 0.5$

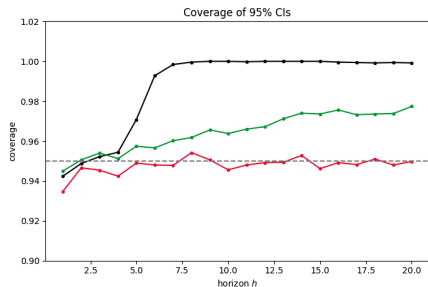


Figure 8: Baseline

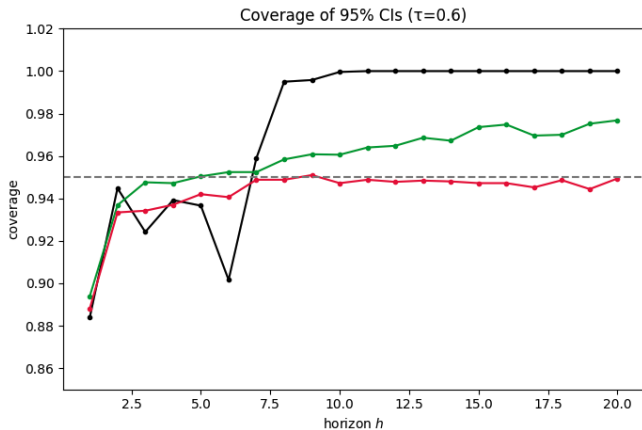


Figure 15: Fct. Misspecification at $\gamma = 0.6$

Discussion of Results

- $LP(4) \approx VAR(15)$
- As T decreases, $LP(4)$ is equal to a less parsimonious VAR $\rightarrow$ relatively better
- As ρ increases, $LP(4)$ is equal to a less parsimonious VAR $\rightarrow$ relatively better
- As τ increases, $LP(4)$ is equal to a less parsimonious VAR $\rightarrow$ relatively better
- At equal lags, $VAR(p)$ is preferable choice if applicable, except for coverage
- higher order VARs closes coverage gap, at cost of bias and variance

Conclusion

1. Ludwig identity largely consistent with MC findings
- 2.

Thank you for your Attention! Questions?